Leeds Hiking Companion API

A GeoDjango-based REST API designed to provide hiking trails, car parks, and transport links for the Leeds and Peak District area. This project features geospatial data ingestion from OpenStreetMap, nearest-neighbor calculation for amenities, and a fully functional review system with authentication.

This API requires **Basic Authentication** - this can be accessed with superuser account Marker with password COMP3011Marker.

Common Error Codes

These errors can occur on any endpoint that requires the `Marker` superuser account.

- **401 Unauthorized**: The request lacks valid authentication credentials. Ensure the `Authorization` header is set to `Basic Auth`.
- **403 Forbidden**: You are authenticated but do not have permission to access the resource (e.g., trying to edit a review you did not create).

These occur when querying for hiking routes or geospatial data.

- **404 Not Found**: The specified `trail` ID does not exist in the database. Replace the ID in the URL with a valid integer.
- **400 Bad Request (Filtered Search)**: An invalid query parameter was provided, such as a difficulty level not listed in the options (`Easy` , `Moderate` , etc.).

These occur during the creation, update, or deletion of user content.

- **400 Bad Request**: This is triggered by validation failures in the request body:
  - **Missing Fields**: Required fields like `rating` or `trail` were not provided.
  - **Invalid Data**: The `rating` was outside the 1–5 range, or the `date_hiked` was not in `YYYY-MM-DD` format.
- **404 Not Found**: Attempting to `GET` , `PUT` , or `DELETE` a review or log entry ID that does not exist or has already been removed.
- **403 Forbidden (Ownership)**: Occurs when a user attempts to `PUT` (update) or `DELETE` a review that belongs to another user

These occur when fetching amenities like car parks or transport links.

- **404 Not Found**: The specific `carpark` or `transport` ID provided in the URL was not found.
- **503 Service Unavailable**: (Internal logic) Occurs if the external weather API (Open-Meteo) or OpenStreetMap data ingestion fails during a real-time request.

AUTHORIZATION Basic Auth

Username	<username>
Password	{{Marker Password}}

AUTHORIZATION Basic Auth

This folder is using Basic Auth from collection Leeds Hiking Companion API

GET GET All Trails

https://joshbellouol.eu.pythonanywhere.com/api/trails/

Retrieves a paginated list of all hiking trails.

- Can be used to display all the trails available

A successful request will return `200 OK`, which is common across all GET requests.

AUTHORIZATION Basic Auth

This request is using Basic Auth from collection Leeds Hiking Companion API

HEADERS

Accept	application/json

Example Request

GET All Trails

curl

```
curl -L 'https://joshbellouol.eu.pythonanywhere.com/api/trails/' \
-H 'Accept: application/json' \
-H 'Authorization: Basic <credentials>'
```

Example Response

Body Headers (13)

200 OK

```json
[
  {
    "id": 1,
    "name": "Upper Don Trail",
    "region": "Peak District",
    "difficulty": "Easy",
    "length": 3.46,
    "elevation_gain": 13,
    "popularity": 0,
    "path": "SRID=4326;MULTILINESTRING ((-1 View More 3921079, -1.4774134 53.3920231, -1.4774685
53.3919181, -1.4774842 53.391786, -1.4770443 53.3911242, -1.4766084 53.3907923, -1.4763255
```


GET GET Filtered Trails

https://joshbellouol.eu.pythonanywhere.com/api/trails/?difficulty=Moderate

Filters trails by difficulty level.

Options: `Easy`, `Moderate`, `Challenging`, `Hard`, `Multi-day Trek`

> Note: Parameter values are case-sensitive. You can combine this with the `search` parameter to find specific trails within a difficulty tier.

AUTHORIZATION Basic Auth

This request is using Basic Auth from collection Leeds Hiking Companion API

PARAMS

difficulty	Moderate

Example Request

GET Filtered Trails

curl

```curl
curl -L 'https://joshbellouol.eu.pythonanywhere.com/api/trails/?difficulty=Moderate'
```

Example Response

Body Headers (14) 200 OK

Text

```
[
  {
    "id": 9,
    "name": "A Walk to the Forest",
    "region": "Peak District",
    "difficulty": "Moderate",
    "length": 8.25,
    "elevation_gain": 234,
    "popularity": 0,
```


View More

GET GET Trail Search

https://joshbellouol.eu.pythonanywhere.com/api/trails/?search=Reservoir

Performs a partial match search on the trail **Name** and **Region**.

AUTHORIZATION Basic Auth

This request is using Basic Auth from collection Leeds Hiking Companion API

PARAMS

search	Reservoir

Example Request

GET Trail Search

curl

```
curl -L 'https://joshbellouol.eu.pythonanywhere.com/api/trails/?search=Reservoir'
```

Example Response

Body Headers (14) 200 OK

Text


```
[
  {
    "id": 52,
    "name": "Pennine Way (Standedge to Warland Reservoir)",
    "region": "Peak District",
    "difficulty": "Moderate",
    "length": 12.77,
    "elevation_gain": 86,
    "popularity": 0,
```

GET GET Trail Detail

https://joshbellouol.eu.pythonanywhere.com/api/trails/1

Retrieves detailed information for a specific trail, including:

- **Geometry:** (GeoJSON format)
- **Linked Services:** Nearest car parks and transport links (automatically calculated).

Options: Replace '1' with any valid trail ID.

AUTHORIZATION Basic Auth

This request is using Basic Auth from collection Leeds Hiking Companion API

Example Request

GET Trail Detail

curl

```curl
curl -L 'https://joshbellouol.eu.pythonanywhere.com/api/trails/1' \
-H 'Authorization: Basic <credentials>'
```

Example Response

Body Headers (13)

200 OK

Text


```text
{
  "id": 1,
  "name": "Upper Don Trail",
  "region": "Peak District",
  "difficulty": "Easy",
  "length": 3.46,
  "elevation_gain": 13,
  "popularity": 0,
  "path": "SRID=4326;MULTILINESTRING ((-1.4773655 53.3921079, -1.4774134 53.3920231, -1.4774685
53.3919181, -1.4774842 53.391786, -1.47704 View More , -1.4766084 53.3907923, -1.4763255
53.3906603, -1.4759834 53.3905651, -1.4759429 53.3905577, -1.4759066 53.3905576, -1.4758785
```


Reviews

AUTHORIZATION Basic Auth

POSTMAN

Documentation Settings

GET GET All Reviews

https://joshbellouol.eu.pythonanywhere.com/api/reviews

Retrieves a paginated list of all hiking trail reviews.

A successful request will return `200 OK`, which is common across all GET requests.

AUTHORIZATION Basic Auth

This request is using Basic Auth from collection Leeds Hiking Companion API

Example Request

GET All Reviews

curl

```
curl -L 'https://joshbellouol.eu.pythonanywhere.com/api/reviews' \
-H 'Authorization: Basic <credentials>'
```

Example Response

Body Headers (13) 200 OK

Text


```
[
  {
    "id": 1,
    "title": "Postman Test",
    "trail": 21,
    "user": "joshb",
    "rating": 5,
    "content": "Security test from Postman!",
    "created_on": "2026-02-11T22:51:00.994037Z"
  },
```


View More

POST POST New Review

https://joshbellouol.eu.pythonanywhere.com/api/reviews/

Allows an authenticated user to submit a new rating and comment for a specific trail.

- **trail** (Integer, Required): The unique ID of the trail being reviewed (e.g., 1).
- **title** (String, Required): A short, descriptive title for your review.

- **active** (Boolean, Optional): Set to `true` by default to make the review visible to others.

> **Note:** Users can only review trails that exist. The `user` field is automatically set to the currently logged-in account.

A successful request will return `201 Created`.

AUTHORIZATION Basic Auth

This request is using Basic Auth from collection Leeds Hiking Companion API

HEADERS

Content-Type	application/json

Body raw (json)

```
{
  "title": "Trail Review",
  "trail": 1,
  "rating": 5,
  "content": "Absolutely stunning views! The path was a bit muddy near the start, but it cleared up. Great for dogs.",
  "active": true
}
```

Example Request

POST New Review

```curl
curl -L 'https://joshbellouol.eu.pythonanywhere.com/api/reviews/' \
-H 'Content-Type: application/json' \
-H 'Authorization: Basic <credentials>' \
-d '{
  "title": "Trail Review",
  "trail": 1,
  "rating": 5,
  "content": "Absolutely stunning views! The path was a bit muddy near the start, but it cleared up. Great for dogs.",
  "active": true
}'
```

Example Response

Text


```
{
  "id": 10,
  "title": "Trail Review",
  "trail": 1,
  "user": "Marker",
  "rating": 5,
  "content": "Absolutely stunning views! The path was a bit muddy near the start, but it cleared up. Great for dogs.",
  "created_on": "2026-03-11T13:07:51.530207Z"
}
```


PUT PUT Review

https://joshbellouol.eu.pythonanywhere.com/api/reviews/7/

Updates the content of an existing review. You must be the **original author** to perform this action.

- **trail** (Integer, Required): The ID of the trail associated with this review.
- **title** (String, Required): Updated title for the review.
- **rating** (Integer, Required): Updated score (1-5).
- **content** (String, Required): Updated review text.
- **active** (Boolean, Required): Status of the review.

> Note: As this is a PUT request, all fields must be sent even if they aren't being changed. Use PATCH for partial updates.

AUTHORIZATION Basic Auth

This request is using Basic Auth from collection Leeds Hiking Companion API

HEADERS

Content-Type	application/json

Body raw (json)

json

```
{
  "title": "Trail Review",
  "trail": 1,
  "rating": 3,
```

```
  "active": true
}
```

Example Request

PUT Review

curl

```curl
curl -L -X PUT 'https://joshbellouol.eu.pythonanywhere.com/api/reviews/7/' \
-H 'Content-Type: application/json' \
-H 'Authorization: Basic <credentials>' \
-d '{
  "title": "Trail Review",
  "trail": 1,
  "rating": 3,
  "content": "Changed rating to 3. Was very muddy.",
  "active": true
}'
```

Example Response

Body Headers (12) 200 OK

Text

```text
{
  "id": 7,
  "title": "Trail Review",
  "trail": 1,
  "user": "joshb",
  "rating": 3,
  "content": "Changed rating to 3. Was very muddy.",
  "created_on": "2026-02-11T22:54:39.695788Z"
}
```

DELETE DELETE Review

https://joshbellouol.eu.pythonanywhere.com/api/reviews/99/

Remove the review of the review ID selected, e.g., /99/

- **Response Code:** `204 No Content`
- **Body:** *(Empty)*
- The review was successfully deleted. No content is returned.
- Returns `404 Not Found` if attempts to delete a non-existent review.

This request is using Basic Auth from collection Leeds Hiking Companion API

Example Request

DELETE Review

curl

```
curl -L -X DELETE 'https://joshbellouol.eu.pythonanywhere.com/api/reviews/99/' \
-H 'Authorization: Basic <credentials>'
```

Example Response

Body Headers (11) 204 NO CONTENT

No response body

This request doesn't return any response body

Services

AUTHORIZATION Basic Auth

This folder is using Basic Auth from collection Leeds Hiking Companion API

GET GET All Car Parks

https://joshbellouol.eu.pythonanywhere.com/api/carparks

Lists all imported car parks within the Leeds/Peak District area.

> **Note:** Fields like `capacity` and `is_free` may be `null` if the data was not available from OpenStreetMap.

AUTHORIZATION Basic Auth

This request is using Basic Auth from collection Leeds Hiking Companion API

Example Request

GET All Car Parks

curl

```
curl -L 'https://joshbellouol.eu.pythonanywhere.com/api/carparks' \
-H 'Authorization: Basic <credentials>'
```

Example Response

Body Headers (13) 200 OK

Text


```
[
  {
    "id": 1,
    "trail": 115,
    "name": "Unnamed Car Park",
    "capacity": null,
    "is_free": null,
    "latitude": "53.742670",
    "longitude": "-2.024899",
    "has_disabled_parking": null
```


View More

GET GET Car Park Detail

https://joshbellouol.eu.pythonanywhere.com/api/carparks/1

Retrieves detailed information for a specific Car Park, including:

- **Capacity**
- Whether you need to pay or not
- Locational data

Options: Replace '1' with any valid Car Park ID.

AUTHORIZATION Basic Auth

This request is using Basic Auth from collection Leeds Hiking Companion API

Example Request

GET Car Park Detail

curl

```
curl -L 'https://joshbellouol.eu.pythonanywhere.com/api/carparks/1' \
-H 'Authorization: Basic <credentials>'
```

Body Headers (12) 200 OK

Text


```
{
  "id": 1,
  "trail": 115,
  "name": "Unnamed Car Park",
  "capacity": null,
  "is_free": null,
  "latitude": "53.742670",
  "longitude": "-2.024899",
  "has_disabled_parking": null
}
```


GET GET All Transport Links

https://joshbellouol.eu.pythonanywhere.com/api/transport

Retrieves a paginated list of all transport links that are close to any trail.

AUTHORIZATION Basic Auth

This request is using Basic Auth from collection Leeds Hiking Companion API

Example Request

GET All Transport Links

curl

```curl
curl -L 'https://joshbellouol.eu.pythonanywhere.com/api/transport' \
-H 'Authorization: Basic <credentials>'
```

Example Response

Body Headers (13) 200 OK

Text


```
[
  {
    "id": 1,
    "trail": 21,
    "name": "Oxenhope",
    "type": "Train",
    "latitude": "53.814922",
```

GET GET Transport Details

https://joshbellouol.eu.pythonanywhere.com/api/transport/1

Retrieves detailed information for a specific Transport Link, including:

- **Type**
- Linked Trail

Options: Replace '1' with any valid Transport Link ID.

AUTHORIZATION Basic Auth

This request is using Basic Auth from collection Leeds Hiking Companion API

Example Request

GET Transport Details

curl

```curl
curl -L 'https://joshbellouol.eu.pythonanywhere.com/api/transport/1' \
-H 'Authorization: Basic <credentials>'
```

Example Response

Body Headers (12) 200 OK

Text

```text
{
  "id": 1,
  "trail": 21,
  "name": "Oxenhope",
  "type": "Train",
  "latitude": "53.814922",
  "longitude": "-1.952482"
}
```

GET GET Free Car Parking

https://joshbellouol.eu.pythonanywhere.com/api/carparks/?is_free=true

Filters the list to show **only free car parks**

- `is_free` : `true` (Free), `false` (Paid), or exclude to see all.

AUTHORIZATION Basic Auth

This request is using Basic Auth from collection Leeds Hiking Companion API

PARAMS

is_free	true

Example Request

GET Free Car Parking

curl

```curl
curl -L 'https://joshbellouol.eu.pythonanywhere.com/api/carparks/?is_free=true' \
-H 'Authorization: Basic <credentials>'
```

Example Response

Body Headers (13) 200 OK

Text


```text
[
  {
    "id": 9,
    "trail": 4,
    "name": "Unnamed Car Park",
    "capacity": 1,
    "is_free": true,
    "latitude": "53.235963",
    "longitude": "-1.436282",
    "has_disabled_parking": null
```


View More

GET GET Parking with Disabled Access

https://joshbellouol.eu.pythonanywhere.com/api/carparks/?has_disabled_parking=true

Returns a list of car parks that have explicitly confirmed disabled parking spaces.

> *Note:* Car parks with unknown status (`null`) are excluded from these results.

AUTHORIZATION Basic Auth

PARAMS

has_disabled_parking	true

Example Request

GET Parking with Disable...

curl

```
curl -L 'https://joshbellouol.eu.pythonanywhere.com/api/carparks/?has_disabled_parking=true'
```

Example Response

Body Headers (14) 200 OK

Text


```
[
  {
    "id": 40,
    "trail": 42,
    "name": "Unnamed Car Park",
    "capacity": 4,
    "is_free": null,
    "latitude": "53.379588",
    "longitude": "-1.457772",
    "has_disabled_parking": true
  },
```


View More

GET GET Car Parks for Specified Trail

https://joshbellouol.eu.pythonanywhere.com/api/carparks/?trail=3

Retrieves all car parks associated with a specific trail (e.g., Trail ID `3`).
Useful for finding parking options near a specific hiking route.

AUTHORIZATION Basic Auth

This request is using Basic Auth from collection Leeds Hiking Companion API

PARAMS

trail	3

Example Request

GET Car Park for Specifie...

curl

```
curl -L 'https://joshbellouol.eu.pythonanywhere.com/api/carparks/?trail=3'
```

Example Response

Body Headers (12) 200 OK

Text


```
[
  {
    "id": 86,
    "trail": 3,
    "name": "Unnamed Car Park",
    "capacity": null,
    "is_free": null,
    "latitude": "53.174135",
    "longitude": "-1.361274",
    "has_disabled_parking": null
  }
```


View More

Trail Log Book

AUTHORIZATION Basic Auth

This folder is using Basic Auth from collection Leeds Hiking Companion API

GET GET All Trail Logbooks

https://joshbellouol.eu.pythonanywhere.com/api/logbook/

Retrieves a paginated list of all logged trails of the currently signed in user.

Endpoint strictly returns logs associated with the authenticated `Basic Auth` user. If no logs exist for the current user, an empty list is returned. You cannot access or modify the logbook entries of other hikers.

AUTHORIZATION Basic Auth

This request is using Basic Auth from collection Leeds Hiking Companion API

Example Request

GET All Trail Logbooks

curl

```
curl -L 'https://joshbellouol.eu.pythonanywhere.com/api/logbook/' \
-H 'Authorization: Basic <credentials>'
```

Example Response

Body Headers (13) 200 OK

Text

```
[
  {
    "id": 4,
    "trail": 99,
    "trail_name": "Peak District Boundary Walk Stage 4",
    "user": "Marker",
    "date_hiked": "2023-10-25",
    "duration_minutes": 120,
    "weather": "Sunny",
    "notes": "Great walk, but forgot my
```

View More

```
e."
  },
```

POST POST New Trail Log

https://joshbellouol.eu.pythonanywhere.com/api/logbook/

Allows an authenticated user to record a personal log entry for a completed hike. These entries are **private**.

- **trail** (Integer, Required): The unique ID of the trail hiked (e.g., 99).
- **date_hiked** (String, Required): The date of the hike in **YYYY-MM-DD** format.
- **duration_minutes** (Integer, Required): Total time taken for the hike in minutes.
- **weather** (String, Required): The weather conditions during the hike. **Options:** `Sunny`, `Rainy`, `Cloudy`, `Snow`.
- **notes** (String, Optional): Private notes about your experience or fitness stats.

> **Note:** Unlike reviews, these logs are **private** and only visible to the user who created them. This is useful for tracking personal fitness stats (duration/frequency).

A successful request will return `201 Created`.

AUTHORIZATION Basic Auth

This request is using Basic Auth from collection Leeds Hiking Companion API

HEADERS

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Body raw (json)

```
{
  "trail": 99,
  "date_hiked": "2023-10-25",
  "duration_minutes": 120,
  "weather": "Sunny",
  "notes": "Great walk, but forgot my water bottle."
}
```

Example Request

POST New Trail Log

```curl
curl -L 'https://joshbellouol.eu.pythonanywhere.com/api/logbook/' \
-H 'Content-Type: application/json' \
-H 'Authorization: Basic <credentials>' \
-d '{
  "trail": 99,
  "date_hiked": "2023-10-25",
  "duration_minutes": 120,
  "weather": "Sunny",
  "notes": "Great walk, but forgot my water bottle."
}'
```

Example Response

Body Headers (12)

201 CREATED

```text
{
  "id": 6,
  "trail": 99,
  "trail_name": "Peak District Boundary Walk Stage 4",
  "user": "Marker",
  "date_hiked": "2023-10-25",
  "duration_minutes": 120,
  "weather": "Sunny",
  "notes": "Great walk, but forgot my water bottle."
}
```

DELETE DELETE Trail Log

Delete selected logbook entry. Entry is selected with the ID, e.g., /5/

- **Response Code:** `204 No Content`
- **Body:** *(Empty)*
- The review was successfully deleted. No content is returned.
- Returns 404 Not Found if attempts to delete a non-existent entry

AUTHORIZATION Basic Auth

This request is using Basic Auth from collection Leeds Hiking Companion API

Example Request

DELETE Trail Log

curl

```
curl -L -X DELETE 'https://joshbellouol.eu.pythonanywhere.com/api/logbook/5/' \
-H 'Authorization: Basic <credentials>'
```

Example Response

Body Headers (11)

204 NO CONTENT

No response body

This request doesn't return any response body